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Green AI Orchestration Bridging Trustworthy AI and Edge AI through tinyML for Frugal Intelligence

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***Abstract*—The rapid proliferation of edge AI, particularly within the TinyML ecosystem, demands innovative approaches to balance computational efficiency, energy consumption, and model performance. This paper explores the relevance of Green AI orchestration—exemplified by initiatives such as the RAIDO project—to the TinyML community, where resource constraints and sustainability are paramount. By integrating energy-efficient hardware, optimized algorithms, and adaptive orchestration frameworks, green AI paradigms allied to TinyML offer a pathway to deploy trustworthy, low-footprint AI models on resource-limited devices. We examine how these principles align with the ethos of frugal AI, which prioritizes minimalism, accessibility, and environmental responsibility. The paper sets the stage for a new era of Frugal Edge AI, where green TinyML orchestration not only reduces operational costs and carbon footprints but also democratizes AI deployment in edge environments. Our discussion underscores the synergy between green, frugal and edge AI, advocating for a holistic approach to AI development that harmonizes technological advancement with environmental stewardship.**

***Index Terms*—Edge Artificial Intelligence, AIoT, Green AI, TinyML, AI Orchestration, Frugal AI, Edge Intelligence, Energy Efficiency, Trustworthy AI, Sustainable Computing**

I. Introduction

The proliferation of Edge AI—where AI models are deployed directly on end devices rather than in centralized data centers—has revolutionized real-time, privacy-preserving, and low-latency applications. However, this paradigm shift introduces significant challenges: edge devices are often constrained by limited computational power, memory, and energy resources. As highlighted in recent literature [2], the success of Edge AI hinges not only on technical efficiency but also on sustainability and accessibility—principles that lie at the heart of the emerging frugal AI movement. Recent advancements in Edge AI have focused on model compression and distributed learning, including methods such as pruning, quantization, and federated learning, to reduce computational demands and energy consumption on edge devices. Along the same lines, RAIDO [8], an EU-funded project, proposes a novel approach focusing on the untaped potential of green AI orchestration, in which an adaptive orchestrator dynamically optimises the entire life cycle of a Machine Learning process across distributed and potentially heterogeneous, resource-constrained resources. In terms of architecture, RAIDO consists of several cooperating layers, including data pre-processing and data generation layers, a model optimisation layer, a green-AI orchestration, and a feedback and monitoring layer, among others. Across these layers, RAIDO can monitor a model's energy use, schedule workflows based on energy consumption, and promote ethical compatibility through a human-in-the-loop or blockchain operating in centralised, distributed, or hybrid deployments.

Meanwhile, the frugal AI literature has expanded the conversation beyond technical efficiency, emphasizing environmental impact, cost-effectiveness, and democratization of AI access. The concept of frugal AI articulated in [3] advocates for AI systems that are resource-efficient, environmentally responsible, and inclusive, prioritizing minimalism without sacrificing performance, with particular impact on sustainable development [6]. This ethos aligns closely with the goals of Edge AI, where the need to operate within strict resource budgets is paramount. Yet, while Edge AI research has made strides in optimizing model architectures, hardware acceleration, and distributed inference [9], the integration of frugal AI principles into Edge AI orchestration remains an open and critical area of inquiry. The concept of *Frugal Edge AI* intersects with several contemporary research trajectories, notably Edge Intelligence, Green AI, and Sustainable Computing. Edge Intelligence has emerged as a response to the latency, privacy, and bandwidth limitations of cloud-based AI systems, focusing on distributing computation closer to data sources [10]. In parallel, Green AI advocates for efficiency and transparency in resource use throughout the AI lifecycle [1], while Sustainable Computing emphasizes energy-aware system design [5]. *Frugal Edge AI* synthesizes these streams by prioritizing algorithmic compactness and energy efficiency through techniques such as model pruning, quantization, and on-device learning, as often exemplified in the field of TinyML [4]. The latter provides a technical bridge between AI and IoT, enabling embedded intelligence in IoT nodes such as sensors, gateways, and actuators. Unlike conventional edge AI, which often assumes moderate hardware capacity and network reliability, *Frugal Edge AI* explicitly targets low-cost, low-power contexts—those typical of rural, remote, or developing regions. Its ambition is not only technical optimization but also social and environmental equity, aligning with broader goals of responsible AI and the Sustainable Development Goals (SDGs). Consequently, *Frugal Edge AI* can be regarded

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as a socio-technical evolution of edge computing—where efficiency, accessibility, and sustainability converge to make intelligence universally deployable.

Despite significant progress in Edge AI and TinyML, current AIoT orchestration approaches lack mechanisms to (i) coordinate computation across highly heterogeneous, low-power edge devices; (ii) dynamically adapt model selection, inference frequency, and workload distribution based on real-time energy availability; and (iii) integrate sustainability constraints—such as energy per inference, carbon intensity, or device wear—into orchestration decisions. Existing edge orchestration architectures typically assume moderate device capabilities and stable connectivity, making them poorly suited for the ultra-low-power microcontroller environments that characterize TinyML deployments. Furthermore, current AIoT pipelines seldom provide unified interfaces for tracking trustworthiness metrics (e.g., privacy, drift, data lineage) alongside energy and computation metrics. This paper addresses these gaps by proposing Frugal Edge AI approach engaging TinyML and Green AI orchestration, positioning RAIDO as a reference architecture capable of delivering adaptive, sustainability-aware orchestration across constrained AIoT systems.

II. Methodology

While Frugal AI broadly emphasizes efficiency, accessibility, and sustainability across the AI lifecycle, TinyML provides a concrete technical setting where these principles can be operationalized at the network's periphery. The emergence of TinyML—machine learning executed directly on ultra-low-power microcontrollers and embedded devices—enables what may be termed Frugal Edge AI. This concept embodies the extension of Frugal AI to the edge computing layer, where intelligence is embedded within inexpensive, energy-efficient, and resource-constrained hardware (see Figure 3 for context on SDG quantification in EdgeAI research). Frugal Edge AI can be defined as an approach to artificial intelligence that applies the principles of frugality—efficiency, affordability, inclusivity, and sustainability—within the context of edge computing. It involves deploying optimized machine learning models, often through TinyML, on low-power and low-cost embedded devices capable of performing on-device inference with minimal energy and computational resources. Frugal Edge AI aims to deliver locally relevant, real-time intelligence without dependence on high-bandwidth connectivity or cloud infrastructure. This paradigm enables scalable and environmentally responsible AI solutions suited to low-resource or remote environments, bridging the digital divide while reducing operational costs and ecological impact. RAIDO's orchestration framework is conceived as a green intelligence fabric that continuously monitors energy, workload, and network conditions across distributed nodes to allocate computational resources dynamically [7]. In the context of TinyML, such orchestration translates into adaptive deployment strategies where models are selectively activated, suspended, or updated according to local energy budgets, sensor activity, and contextual needs [9].

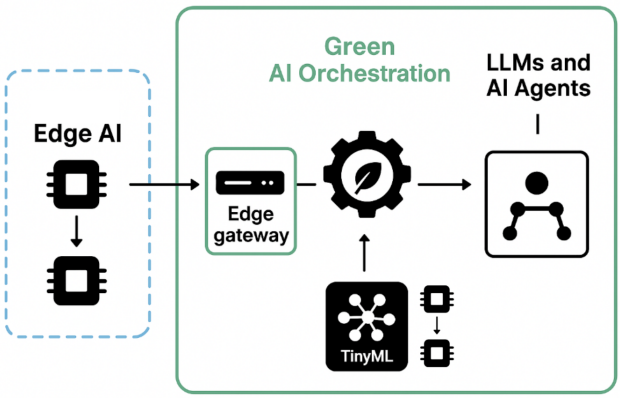


Fig. 1. Overall relation between TinyML and Green AI orchestration as AI agent enabler, independent of Edge gateways when considering the TinyML approach.

The RAIDO orchestration framework enables intelligent coordination of AI workloads across the three tiers of the IoT ecosystem—device, gateway, and cloud—optimizing performance, energy efficiency, and sustainability. At the device layer, TinyML-enabled microcontrollers perform localized inference directly on IoT sensors and embedded systems, enabling real-time decision-making while drastically reducing data transmission and energy consumption. The gateway layer, comprising edge servers, plays a pivotal role as the intermediate intelligence hub: it aggregates and refines device-level outputs, executes medium-complexity analytics, and serves as the most suitable platform for deploying lightweight or distilled LLMs. These edge servers provide the necessary computational capacity to run context-aware, domain-specific LLMs close to data sources—ensuring low latency, enhanced privacy, and reduced dependence on cloud infrastructure. At the cloud layer, RAIDO oversees long-term orchestration, global model retraining, and system-wide optimization, leveraging aggregated insights from the edge to update models and orchestration policies dynamically. This hierarchical design minimizes redundant data flow, balances computational load across tiers, and significantly lowers the overall carbon footprint. Through this multi-tier, LLM-enabled orchestration, RAIDO operationalizes Green AI and Frugal AI principles within IoT ecosystems, creating a sustainable, adaptive, and intelligent continuum from microcontrollers to cloud analytics.

A representative RAIDO use-case is smart-home scenario, where it functions as an energy-aware orchestration layer bridging the home microgrid with the provider's external large-scale forecasting models (see Figure 2. The smart home constantly collects measurements from appliances and transmits them to the energy provider for day-ahead forecasting with the support of heavier models. RAIDO coordinates this interaction by deciding which lightweight tasks run locally on TinyML-capable devices and which require offloading. That ensures efficient operation across the distributed system.

Frugal Edge AI leverages the computational compactness of TinyML to perform real-time inference without relying on con-

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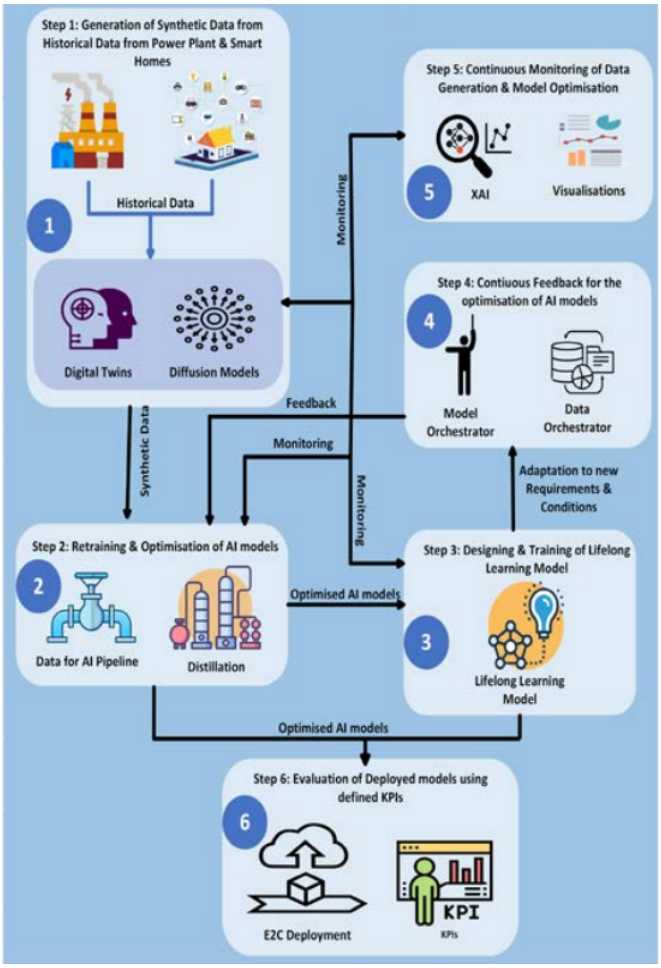


Fig. 2. AI orchestration applying the RAIDO methodology to the context of edge computing in smart home scenarios.

tinuous cloud connectivity or high-end computing infrastructure. Models are pruned, quantized, and optimized to fit within kilobytes of memory, allowing them to run autonomously on microcontrollers powered by minimal energy sources such as batteries or solar cells. This approach directly aligns with the frugal design philosophy of achieving maximum utility with minimal resources. It also supports greater inclusivity by enabling AI deployment in remote or resource-limited environments where network bandwidth, power supply, or financial resources are constrained. TinyML model deployment in our setting follows a constrained pipeline: (i) on-device data collection using sensors (typically sampling at ¡1 kHz); (ii) microcontroller-level feature extraction using CMSIS-DSP or similar libraries; (iii) inference using models quantized to INT8 and compressed below 50–150 kB; (iv) intermittent connectivity for offloading only when required; and (v) AI orchestration deciding when to trigger inference, update the model, or offload computation.Examples of models compatible with this orchestration layer include 1D CNNs for vibration or acoustic monitoring (¡30 kB), 2-layer LSTMs for time-series forecasting (¡80 kB), and keyword spotting networks such as DS-CNN (¡50 kB). To preserve battery, TinyML devices are normally equipped with thin connectivity capabilities such as LoRa. These connectivity technologies are not equally bidirectional, making it important for orchestration strategies to take into account the very limited bandwidth and limited transmission windows from the orchestrator to the device. This poses an additional challenge to the orchestrator design Frugal Edge AI represents a paradigm shift toward the sustainable deployment of Agentic AI in environments constrained by energy, computation, and connectivity (as represented in Figure 1 and detailed in Table I where the VADER sentiment is computed to provide the level of confidence in the obtained and published results). By coupling frugality with Green AI orchestration, it becomes possible to manage distributed AI agents in an adaptive, energy-aware, and context-sensitive manner. The orchestration layer continuously monitors system parameters, dynamically allocating workloads, selecting lightweight TinyML models, and optimizing inference frequency in response to changing environmental and operational conditions. This coordination ensures that AI agents remain effective even under severe resource limitations, extending device longevity and reducing carbon and material footprints. Beyond technical efficiency, this approach democratizes access to intelligent systems, enabling local industries, small enterprises, and community initiatives in regions like rural India or sub-Saharan Africa to harness AI for tasks such as precision agriculture, environmental monitoring, and public health surveillance. In doing so, Frugal Edge AI orchestrated through green governance frameworks not only reduces ecological impact but also fosters inclusive and sustainable digital economies—embedding environmental stewardship and social equity at the core of AI deployment.

TABLE I
THE AMOUNT OF PUBLICATIONS OVER THE TOP 10 SDGS DEPICTED ON THE EDGE AI RESEARCH AREAS AND THEIR COMPUTED AVERAGE SENTIMENT

SDG	Count	Sentiment
9: Industry Innovation	1,927	0.935
3: Health-Well being	1,153	0.933
8: Economic Growth	583	0.953
4: Education	569	0.913
2: Food Security	555	0.96
15: Life on Land	502	0.931
12: Responsible Consumption	280	0.92
13: Climate Crisis	264	0.932
11: Resillient Communities	261	0.943
6: Water Sustainability	245	0.542

III. EARLY CONTRIBUTIONS TO FRUGAL EDGE AI FROM TINYML CASE STUDIES

The Frugal Edge AI paradigm advances sustainability goals by reducing data transmission and energy consumption, which are significant contributors to AI's carbon footprint. By processing data locally, it minimizes dependency on energy-intensive data centers and ensures data privacy—critical for sensitive applications in health, environment, and governance. In regions such as India and across several African countries,

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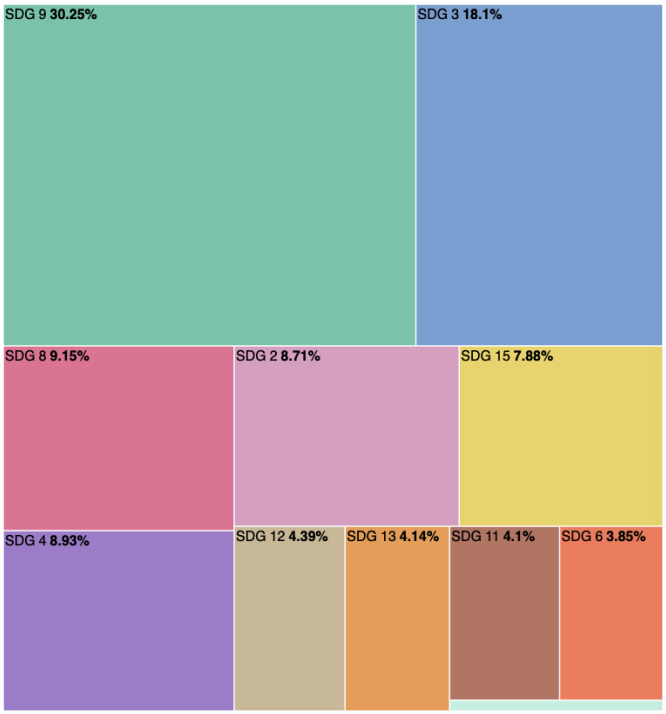


Fig. 3. The SDG quantification in current EdgeAI research

this model has already demonstrated tangible benefits through low-cost deployments for agricultural monitoring, air-quality sensing, and wildlife protection. The concept of Frugal Edge AI contributes a crucial dimension to the broader discourse on AI for Sustainable Development. By embedding intelligence within low-cost, low-power devices, it enables scalable digital innovation that remains aligned with ecological and social constraints. Moreover, its inclusive and context-sensitive design fosters technological equity, ensuring that AI-driven benefits reach underserved or infrastructure-poor communities. As such, Frugal Edge AI offers both a conceptual and operational framework for advancing sustainable, responsible, and democratized AI ecosystems, reinforcing the alignment between technological advancement and global sustainability objectives.

In rural India, for instance, TinyML-enabled smart agriculture nodes are being explored for soil-moisture prediction and pest monitoring in states such as Maharashtra and Andhra Pradesh. These devices—often battery-powered or solar-assisted—operate under strict energy constraints. A RAIDO-like orchestrator can dynamically throttle model complexity or inference frequency depending on environmental conditions: during daytime, when solar energy is abundant, models with richer features may operate, while at night, ultra-compressed variants maintain essential monitoring functions. Through this context-aware adaptation, the system ensures both operational continuity and minimal energy waste. In India's participatory air-quality and water-monitoring projects, low-cost TinyML sensors empower citizens to collect localized data in regions where governmental monitoring networks are sparse. RAIDO's orchestration framework could manage the lifecycle of these distributed devices—ensuring firmware consistency, validating data provenance, and optimizing inference workloads according to regional environmental dynamics. Moreover, RAIDO's explainability modules would allow local communities and policy actors to understand how data are processed and decisions made, reinforcing transparency and civic trust.

Similarly, in sub-Saharan Africa, community-driven climate and biodiversity observatories rely on edge AI for local sensing (e.g., acoustic detection of species, monitoring of illegal logging, or crop stress estimation). RAIDO's orchestration principles enable workload balancing between devices and intermediate gateways: nodes with sufficient battery or connectivity can process heavier inference tasks or aggregate data, while others enter low-power standby modes. This cooperative scheduling not only extends network longevity but also aligns with circular-economy goals by reducing premature device failure and e-waste. In Kenya and Ghana, citizen observatories for biodiversity monitoring (e.g., acoustic bird detection networks) exemplify another sphere where orchestrated frugality becomes impactful. A RAIDO-enabled coordination platform could dynamically schedule sensor activity to coincide with known species migration periods, reduce redundant sensing across overlapping regions, and balance data upload frequency to minimize network congestion. This contextual coordination transforms scattered low-cost devices into a coherent, self-managing environmental intelligence network—an embodiment of participatory sustainable AI in practice.

This integration advances a vision of Frugal-Orchestrated AI—a model of intelligence that is small yet scalable, efficient yet ethical, and local yet globally networked. For developing regions such as India and Africa, where sustainability and accessibility are twin imperatives, such orchestrated TinyML ecosystems could redefine how AI contributes to climate resilience, resource efficiency, and participatory innovation. The integration of RAIDO's green AI orchestration within Frugal Edge AI could unlock transformative possibilities across sectors by enabling intelligent, energy-efficient, and autonomous systems enhanced by AI agents. In smart city IoT ecosystems, it can orchestrate energy-aware traffic and pollution monitoring networks, dynamically balancing workloads among TinyML-based sensors, edge gateways, and AI agents operating at the network's edge. These agents can make localized decisions—such as adjusting traffic-light timings or activating air-quality alerts—without relying on continuous cloud connectivity, ensuring responsiveness and sustainability. In the healthcare IoT domain, wearable devices equipped with TinyML and coordinated through RAIDO's orchestration layer can host personalized AI agents that continuously monitor vital signs, detect anomalies, and deliver privacy-preserving insights directly on-device or through nearby edge servers. Collectively, these cross-sector applications highlight how Frugal Edge AI and RAIDO's orchestration can enable a new generation of self-managing, sustainable AI agent ecosystems that drive innovation in IoT environments while maintaining

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energy efficiency, ethical compliance, and social inclusivity.

IV. Discussion of synergies between Frugal Edge AI, and Green Orchestration

The integration of RAIDO's Green AI orchestration with TinyML-based edge intelligence offers a transformative pathway for deploying resource-efficient, context-aware, and socially inclusive AI in emerging regions. The promise of this synthesis becomes particularly visible when observed through the lens of practical implementations in India and Africa, where resource constraints, infrastructural heterogeneity, and environmental challenges intersect. These geographies—often characterized by intermittent connectivity, limited energy access, and high environmental variability—provide compelling testbeds for evaluating how orchestrated, frugal AI systems can support sustainable development and citizen empowerment.

The deployment and scaling of Frugal Edge AI and Green AI orchestration demand more than technological advancement—they require robust and forward-looking policy ecosystems. Governments and funding agencies play a pivotal role in driving sustainability-oriented AI innovation by incentivizing low-power, resource-efficient design through targeted grants, subsidies, and procurement criteria that reward energy efficiency and reduced lifecycle emissions. Establishing measurable sustainability metrics—such as energy per inference or total carbon cost—can help embed ecological accountability into AI research and development. At the same time, the creation of standards and certification frameworks for TinyML devices and frugal AI models would ensure transparency, interoperability, and public trust in sustainability claims. Equitable access to data and collaborative innovation are equally essential to scaling Frugal Edge AI in low-resource contexts. Open, context-specific datasets for sectors such as agriculture, healthcare, and environmental monitoring enable local adaptation and model frugality, while respecting privacy and data sovereignty. Policymakers should actively foster public–private partnerships and regional innovation ecosystems that connect research institutions, startups, and community organizations. Such collaboration supports the co-design, deployment, and maintenance of sustainable AI solutions that respond to local needs and resource constraints, transforming small-scale prototypes into impactful real-world systems.

Beyond technical efficiency, RAIDO's orchestration layer introduces a governance and transparency dimension essential for sustainable AI ecosystems. Its capacity for traceable decision-making, explainability, and ethical compliance enables AI systems that are not only energy-efficient but also socially trustworthy. This is particularly salient in citizen-science initiatives emerging across India and Africa, where communities engage directly in environmental data collection and stewardship.

Finally, building capacity and embedding ethics and inclusivity into policy frameworks ensures that the benefits of Frugal Edge AI extend to all. Targeted training programs in TinyML, edge computing, and energy-aware AI design can strengthen the local workforce and support sustainable digital transformation in developing regions. Regulatory instruments should further integrate sustainability and social impact into AI procurement and governance, mandating that systems deployed in marginalized communities remain transparent, privacy-preserving, and inclusive (summarised in Table II). By embedding sustainability, equity, and accountability into both policy and practice, governments can establish a foundation for the responsible and enduring use of Frugal Edge AI and green orchestration technologies in achieving global sustainable development goals. By combining technical standards, funding incentives, open data policies, public-private collaboration, sustainable procurement, workforce development, and inclusive governance, policymakers can create an enabling environment for Frugal Edge AI. Such frameworks will support scalable, energy-efficient, and socially responsible AI ecosystems, allowing advanced AI—ranging from TinyML to LLM-enabled agents—to flourish even in low-resource settings, while aligning with SDGs.

TABLE II
Synergies between Frugal Edge AI and Green Orchestration

Scenario	Frugal Edge AI	Green Orchestration	Synergistic Outcome
Energy Efficiency	TinyML models minimize power and data use.	Adaptive scheduling and workload balancing.	Lower carbon footprint, extended device life.
Scalability	Lightweight, low-cost deployment on edge devices.	Hierarchical coordination across device–gateway –cloud.	Efficient scaling in heterogeneous environments.
Governance	Community-driven, transparent sensing.	Traceability and ethical compliance mechanisms.	Trustworthy and accountable AI systems.
Inclusivity	Affordable intelligence for low-resource users.	Policy incentives and open collaboration frameworks.	Broader AI accessibility and digital equity.
Sustainability	Supports SDGs via local, low-power AI.	Monitors lifecycle impact and energy metrics.	Environmentally responsible AI ecosystems.

V. Implications and Future Directions

This paper has examined the emerging paradigm of Frugal Edge AI, demonstrating how TinyML and energy-efficient edge computing operationalize the principles of Frugal AI by enabling intelligent, low-power systems that function effectively in resource-constrained and remote environments. Through localized inference, adaptive design, and minimal infrastructure dependency, Frugal Edge AI minimizes environmental impact while broadening the accessibility of AI technologies. The integration of Green AI orchestration, exemplified by frameworks such as RAIDO, extends this paradigm by coordinating resource-efficient AI agents across heterogeneous networks. This synthesis illustrates that frugality, efficiency, and sustainability can coexist with high performance, forming

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the basis for inclusive, responsible, and context-aware AI ecosystems aligned with global sustainability objectives. Realizing the full potential of Frugal Edge AI, however, requires supportive policy frameworks, standardization, and capacity building. Incentivizing low-power innovation, creating open datasets for local adaptation, and fostering public–private partnerships are critical to ensure that AI advancements remain both equitable and environmentally sound. Future research should focus on adaptive orchestration strategies, context-aware sustainability benchmarks, and real-world validation across sectors such as agriculture, health, and urban infrastructure. As AI continues to evolve toward distributed, autonomous agents, Frugal Edge AI—guided by green orchestration and ethical governance—offers a transformative pathway to deploy intelligence sustainably, bridging technological advancement with social inclusion and environmental stewardship.

Looking forward, several avenues of further research and development are evident. First, exploring adaptive orchestration strategies for dynamic workload allocation and hybrid TinyML/LLM pipelines can optimize energy use and performance across large edge networks. Second, the development of context-aware frugal AI benchmarks and evaluation metrics—including energy-per-inference, carbon footprint, and socio-economic impact—will help quantify and standardize sustainability in AI design. Third, applied studies in real-world domains such as agriculture, health, education, and smart cities are needed to validate the social and economic benefits of Frugal Edge AI. Finally, investigating autonomous, low-cost AI agents capable of cooperative learning, decision-making, and multimodal sensing at the edge will expand the frontier of what frugal, decentralized intelligence can achieve.

Frugal Edge AI represents a transformative approach to Edge AI design and deployment, merging technical innovation with social and environmental responsibility. By embedding intelligence at the edge in an energy- and resource-conscious manner, it promises to extend the reach of AI to underserved populations, advance sustainability objectives, and provide a practical pathway for scaling advanced AI capabilities—including LLMs and AI agents—across diverse global contexts. The evolution of Frugal Edge AI toward large-scale AIoT ecosystems will require unified orchestration frameworks capable of managing millions of interconnected IoT devices with adaptive intelligence. Future research should explore federated and cooperative AIoT architectures that integrate TinyML inference, on-device learning, and green orchestration to achieve scalable, carbon-neutral IoT infrastructures.

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